# Name: Yahya Raees Khan

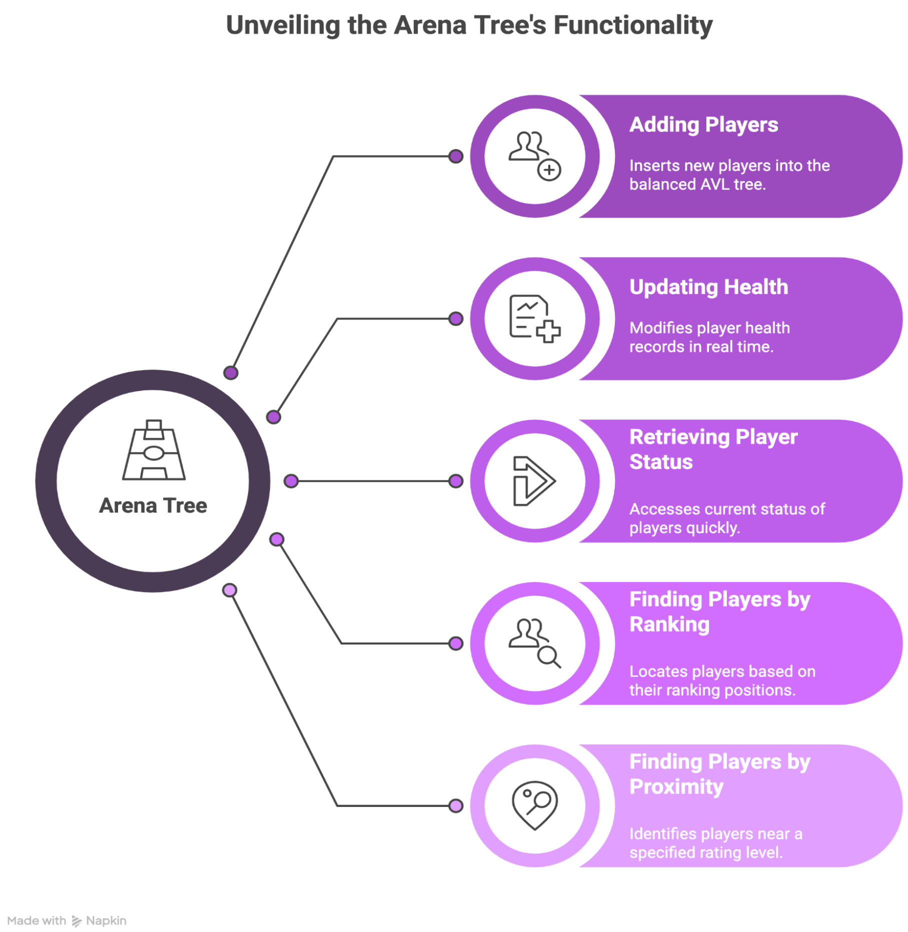
# Program: BSAI

# Group: B

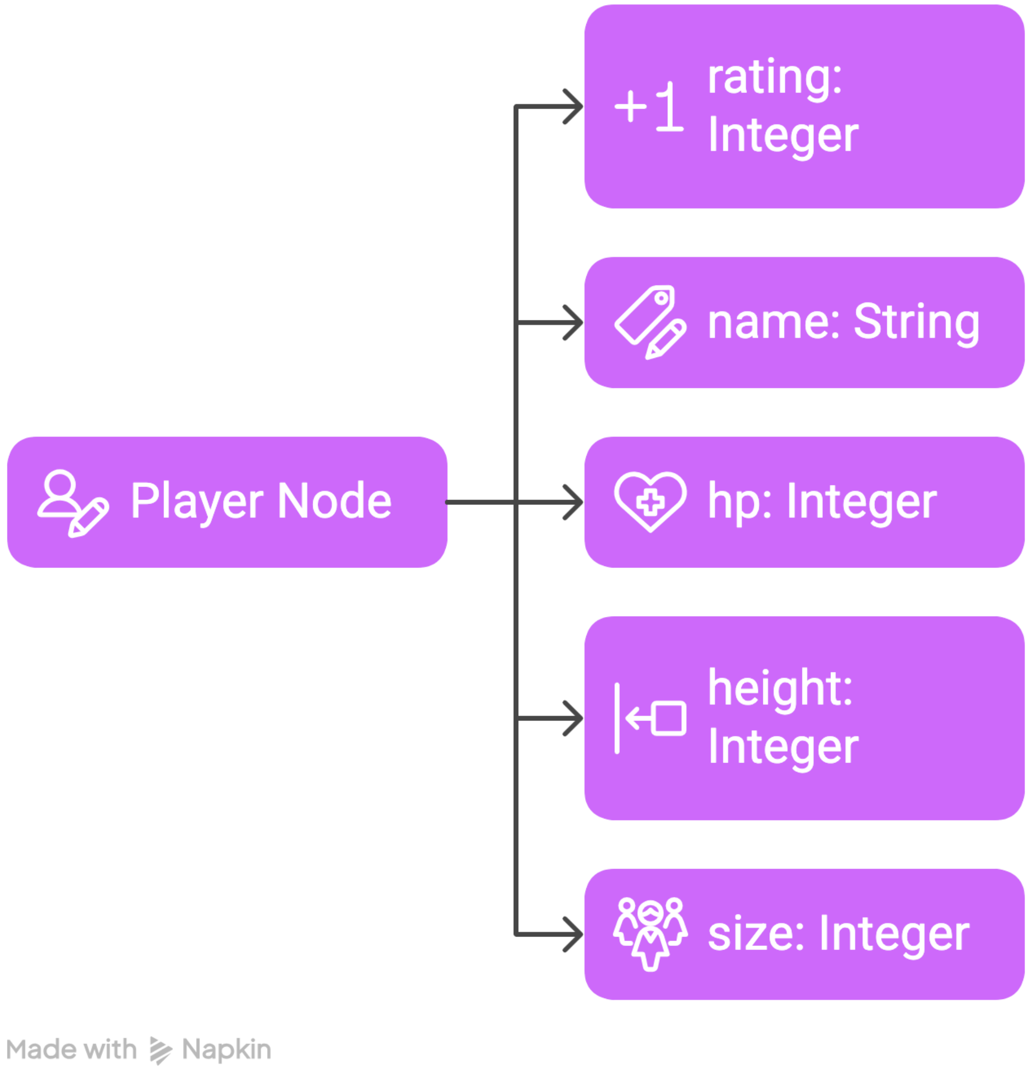
# Subject: Data Structures Algorithms (Lab)

## Code Link

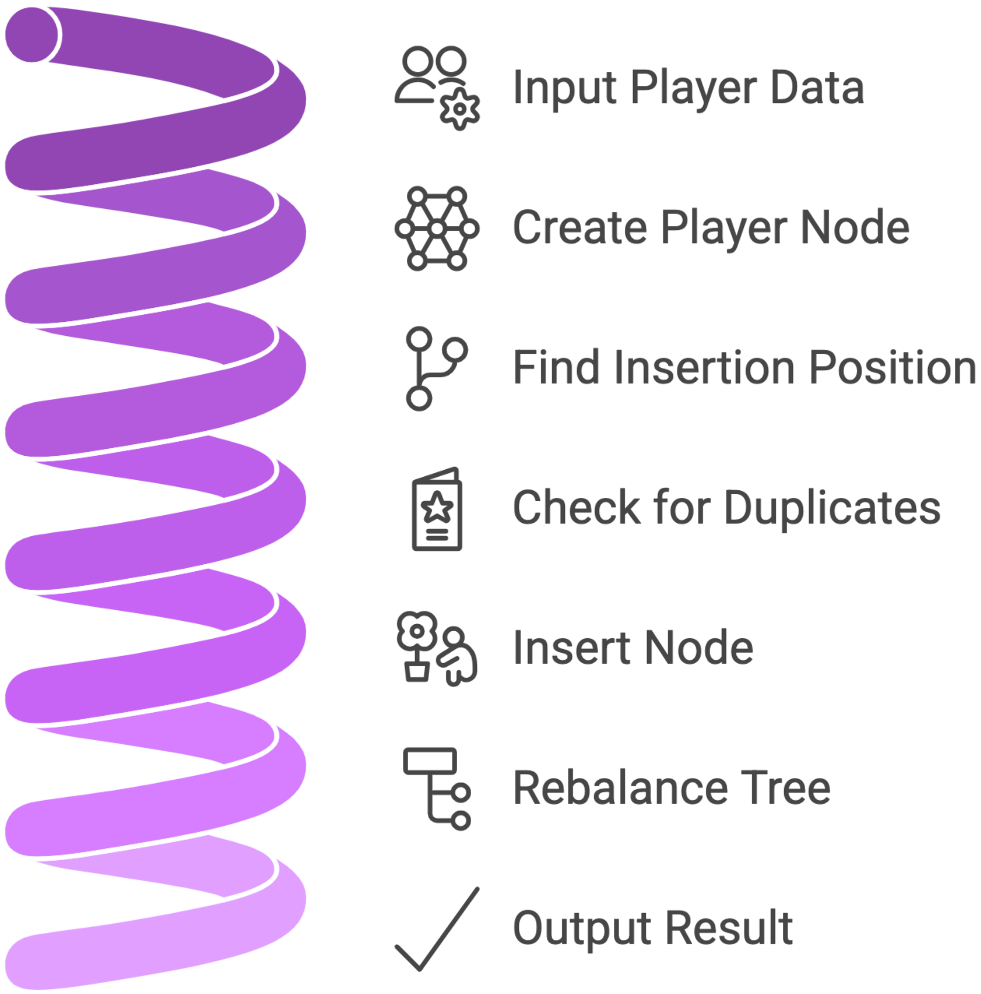
## Visuals:



# Player Data Fields



# Player Insertion Process



# How to remove a player from the Arena Tree?

### Node Has Zero or One Child

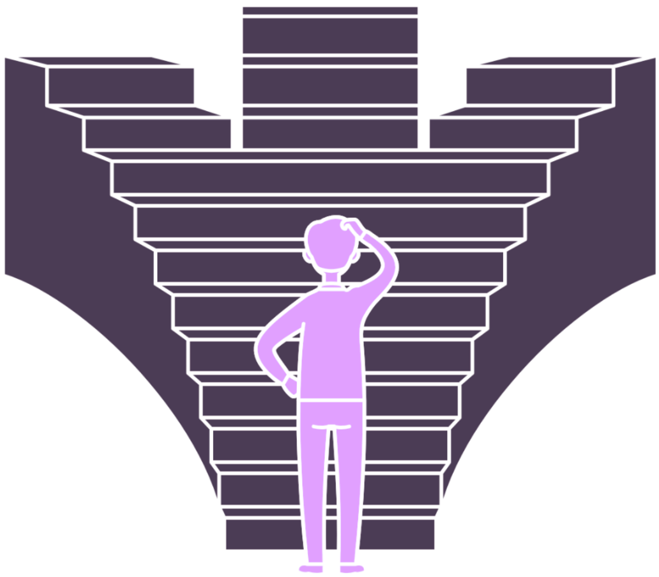
The node can be directly removed without affecting tree balance.

### Node Has Two Children

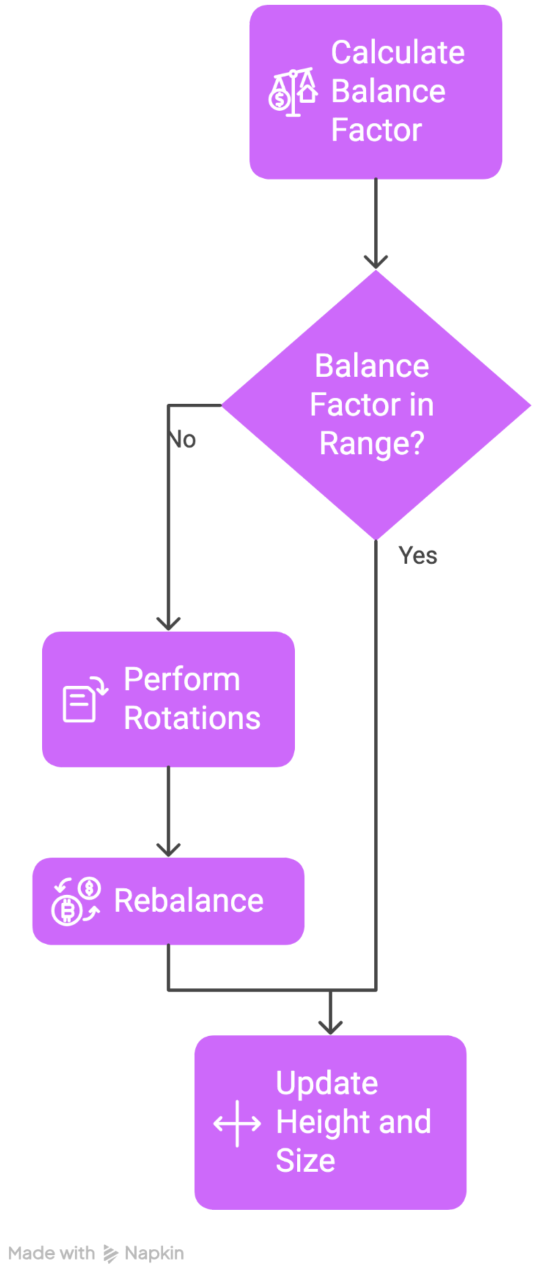
Replace the node's data with the inorder successor's and delete the successor.

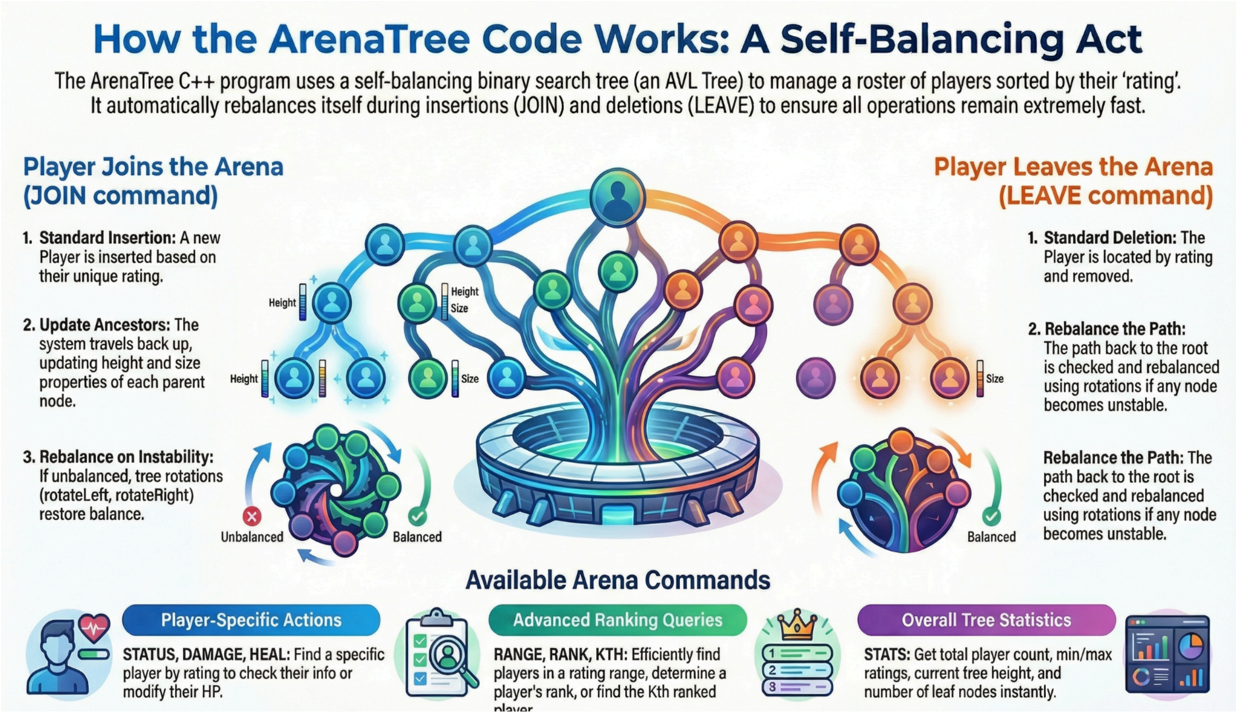
### Player Not Found

Indicate that the player is not in the tree.



## AVL Tree Balancing Process




How the ArenaTree Code Works: A Self-Balancing Act
The ArenaTree C++ program uses a self-balancing binary search tree (an AVL Tree) to manage a roster of players sorted by their 'rating'.
It automatically rebalances itself during insertions (JOIN) and deletions (LEAVE) to ensure all operations remain extremely fast.
Player Joins the Arena (JOIN command)
1. Standard Insertion: A new Player is inserted based on their unique rating.
2. Update Ancestors: The system travels back up, updating height and size properties of each parent node.
3. Rebalance on Instability: If unbalanced, tree rotations (rotateLeft, rotateRight) restore balance.
Height
Size
Unbalanced
Balanced
Player Leaves the Arena (LEAVE command)
1. Standard Deletion: The Player is located by rating and removed.
2. Rebalance the Path: The path back to the root is checked and rebalanced using rotations if any node becomes unstable.
Rebalance the Path: The path back to the root is checked and rebalanced using rotations if any node becomes unstable.
Available Arena Commands
Player-Specific Actions
STATUS, DAMAGE, HEAL: Find a specific player by rating to check their info or modify their HP.
Advanced Ranking Queries
RANGE, RANK, KTH: Efficiently find players in a rating range, determine a player's rank, or find the Kth ranked player.
Overall Tree Statistics
STATS: Get total player count, min/max ratings, current tree height, and number of leaf nodes instantly.

# Thank You 🙌